Machine Learning in Practice I
Lecture 12

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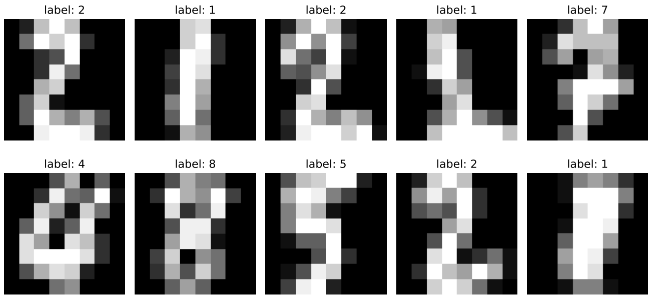
Examples from Computer Vision

Examples from Computer Vision

- After learning the most important ML methods, let us apply them!
- Possible application area: Computer Vision (cv)
- Examples of popular databases:
  - sklearn’s load_digits dataset
  - sklearn’s fetch_lfw_people dataset
  - MNIST dataset

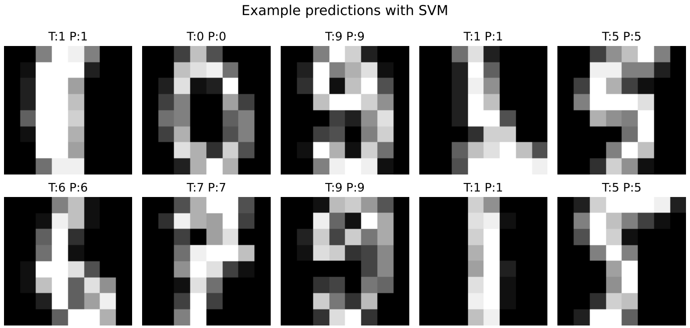
load_digits dataset

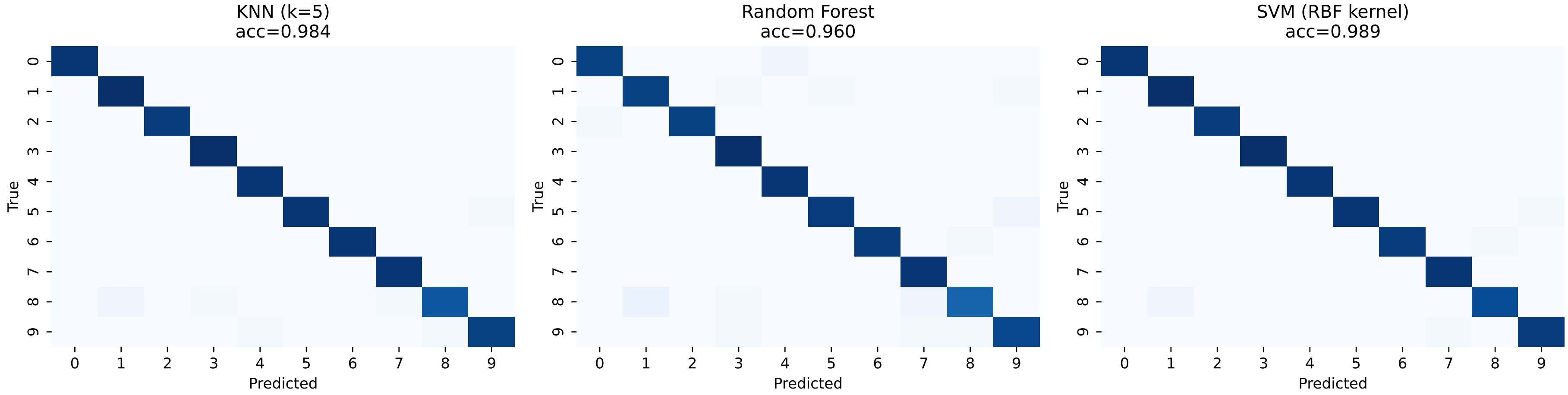
- database properties
  - contains images of handwritten characters: 0-9 (10 classes)
  - contains 1797 sample, 8x8 grayscale (4 bit) images
  - originally collected by the UCI Machine Learning Repository in the 1980s–1990s, from scanned postal code digits and forms
  - one of the first machine learning benchmark studies for optical character recognition (OCR), preceding MNIST
  - in Python available as
    `from sklearn.datasets import load_digits`



load_digits dataset

- basic classifiers
  - train KNN, RF and SVM on the data
  - very good results!





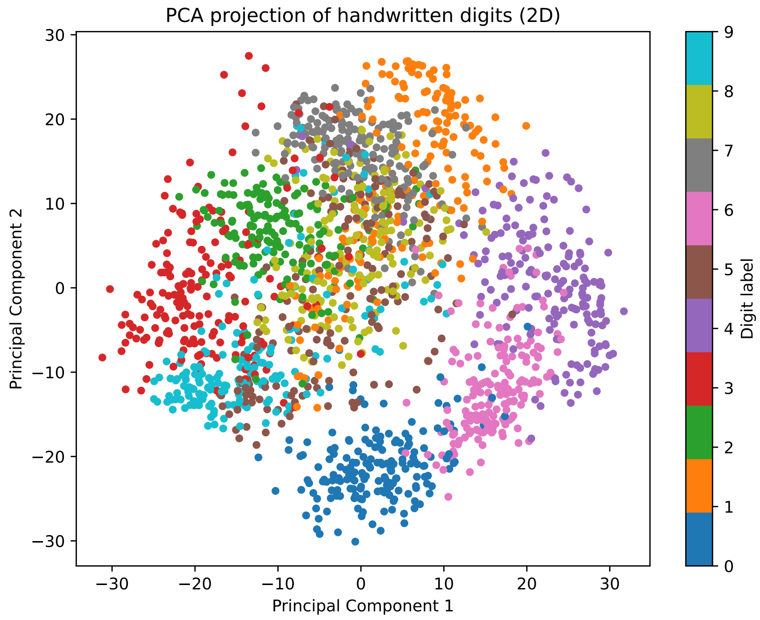
load_digits dataset

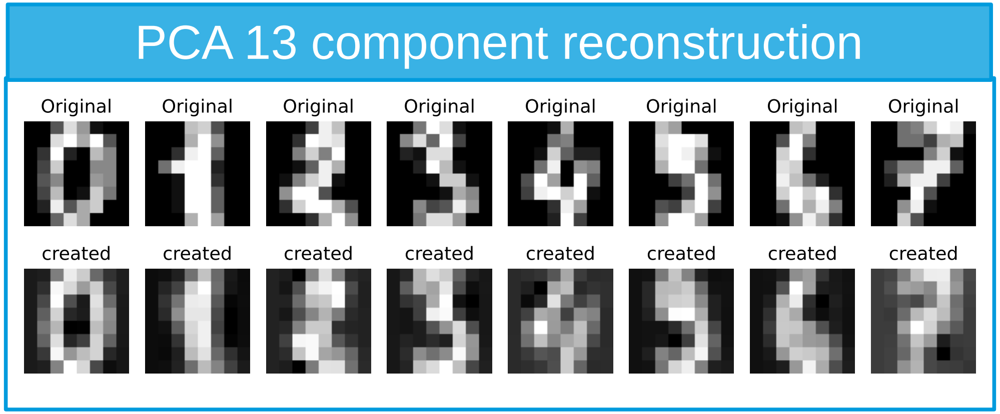
- accuracies:

method	accuracy
DT	82.0%
RF	96.0%
Bagging	92.2%
AdaBoost	82.0%
KNN (k=5)	98.4%
SVM (RBF kernel)	98.9%
NaiveBayes	82.3%
ELM (30k hidden)	98.7%
LLT	98.4%

load_digits dataset

- data compression and reconstruction → PCA





Lessons and conclusions

Conclusions and lessons

- there are a lot of models for few degrees of freedom
- most powerful
  - for classification: RF, KNN, SVM → ELM, LLT
  - compression: PCA
  - unsupervised clusterization: GMM
- for lot of degrees of freedom: lose efficiency
- solution: hierarchical approach, process a small portion per step
  - deep neural networks: translation (CNN) + scaling (pooling)
  - LLMs: attention heads

END OF LECTURE 12